

A Novel Approach to Temporal QoS Estimation via Extended Kalman Filter-Incorporated Latent Feature Analysis

Supplementary File

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I. CONVERGENCE ANALYSIS FOR EKL

1) Implementation of Step 1

Lemma 1: $\varepsilon(p_{(t)u}^k | q_s^{k-1})$ is strongly convex if δ is the minimum singular value for a positive semidefinite matrix.

Proof: Considering user u at time slot t , according to the MFP learning scheme (5)-(9) in the main manuscript, we separate (23a) for user u at time slot t :

$$J(p_{(t)u}^k) = \varepsilon(p_{(t)u}^k | q_s^{k-1}). \quad (S1)$$

Considering the detail expression, we have that:

$$\begin{aligned} J(p_{(t)u}^k) &= \sum_{y_{(t)u,s} \in \Lambda_{(t)u}} \lambda (y_{(t)u,s} - p_{(t)u}(q_s)^T)^2 \\ &= \lambda \|y_{(t)u} - O(p_{(t)u})(M_{(t)u})^T\|_2^2. \end{aligned} \quad (S2)$$

By introducing two arbitrary vectors c and d into (S2) and expanding it to second-order Taylor series, we have:

$$J(c) = J(d) + \frac{\partial J(d)}{\partial d} (c - d)^T + \frac{1}{2} (c - d)^T \frac{\partial^2 J(d)}{\partial d^2} (c - d)^T. \quad (S3)$$

Considering **Definition 4** in the main manuscript, we have that:

$$J(c) \geq J(d) + \frac{\partial J(d)}{\partial d} (c - d)^T + \frac{\delta}{2} \|c - d\|_2^2. \quad (S4)$$

Hence, we need to determine δ such that the following equality holds:

$$(c - d)^T \frac{\partial^2 J(d)}{\partial d^2} (c - d)^T \geq \delta \|c - d\|_2^2. \quad (S5)$$

Since $M_{(t)u}$ is the subset of Q and can be regard as a constant in this step, we simplify the differentiation process by making $\omega(p_{(t)u}) = O(p_{(t)u})(M_{(t)u})^T$. Hence, according to (S2), we have the following inference:

$$\frac{\partial^2 J(b)}{\partial b^2} = 2\lambda \left((\omega'(p_{(t)u}))^2 - (y_{(t)u} - \omega'(p_{(t)u})) \omega''(p_{(t)u})) \right). \quad (S6)$$

where $\omega'(\cdot)$ and $\omega''(\cdot)$ denotes the first-order and second-order derivatives of $\omega(\cdot)$, respectively. Further, we obtain the following inequality based on (S4) and (S5):

$$\begin{aligned} (c - d)^T & \left(2\lambda \left[\frac{(\omega'(p_{(t)u}))^2}{-(y_{(t)u} - \omega'(p_{(t)u})) \omega''(p_{(t)u}))} \right] - \delta I \right) \\ & \cdot (c - d)^T \geq 0 \end{aligned} \quad (S7)$$

which indicates that the matrix $2\lambda((\omega'(p_{(t)u}))^2 - (y_{(t)u} - \omega'(p_{(t)u})) \omega''(p_{(t)u}))) - \delta I$ is positive semidefinite. According to [S1], when δ is the minimum singular value of $2\lambda((\omega'(p_{(t)u}))^2 - (y_{(t)u} - \omega'(p_{(t)u})) \omega''(p_{(t)u})))$, it is positive semidefinite. Hence, *Lemma 1* stands.

After substituting $p_{(t)u}^{k-1}$ and $p_{(t)u}^k$ into (S1), we calculate the gap between $\varepsilon(u_{(t)i}^{k-1} | s_j^{k-1})$ and $\varepsilon(u_{(t)i}^k | s_j^{k-1})$:

$$J(p_{(t)u}^k) - J(p_{(t)u}^{k-1}) = \varepsilon(p_{(t)u}^k | q_s^{k-1}) - \varepsilon(p_{(t)u}^{k-1} | q_s^{k-1}). \quad (S8)$$

Based on *Lemma 1*, the following inferences is derived:

$$J(p_{(t)u}^{k-1}) - J(p_{(t)u}^k) \geq -\frac{\partial \varepsilon(p_{(t)u}^k)}{\partial p_{(t)u}^k} (p_{(t)u}^{k-1} - p_{(t)u}^k)^T. \quad (S9)$$

Based on (5)-(9) in the main manuscript, EKF can be regarded as a recursive ALS algorithm with random disturbance. Consequently, $\partial \varepsilon(p_{(t)u}^k)/\partial p_{(t)u}^k = 0$ is obtained according to a recursive ALS algorithm. Thus, we have

$$\begin{aligned} \forall u \in U, t \in T : J(p_{(t)u}^{k-1}) - J(p_{(t)u}^k) &\geq 0 \\ \Rightarrow \forall u \in U : \sum_{t=1}^{|T|} \varepsilon(p_{(t)u}^k | q_s^{k-1}) &\leq \sum_{t=1}^{|T|} \varepsilon(p_{(t)u}^{k-1} | q_s^{k-1}). \end{aligned} \quad (\text{S10})$$

Hence, **Step 1** is implemented.

2) Implementation of Step 2

Proof: Considering service s , we have the following expression based on (14) in the main manuscript:

$$\begin{aligned} F(q_s) &= \sum_{t=1}^{|T|} \varepsilon(q_s | p_{(t)u}) \\ &= \lambda \|\dot{Y}_s - q_s \dot{P}_s\|^2 + |\Lambda_s| \|q_s\|^2. \end{aligned} \quad (\text{S11})$$

Similar to **Step 1**, it is clear to see that $F(q_s)$ still has strong convexity. Hence, by considering q_s^{k-1} and q_s^k , we have the following equality:

$$F(q_s^{k-1}) - F(q_s^k) \geq \frac{\partial F(q_s^k)}{\partial q_s^k} (q_s^{k-1} - q_s^k)^T. \quad (\text{S12})$$

According to the DFP learning scheme, we have $\partial F(q_s^k)/\partial q_s^k = 0$. Therefore, (S12) can be simplified as follows:

$$F(q_s^k) \leq F(q_s^{k-1}). \quad (\text{S13})$$

Combining (S11) with (S13), we have that:

$$\forall s \in S : \sum_{t=1}^{|T|} \varepsilon(q_s^k | p_{(t)u}^k) \leq \sum_{t=1}^{|T|} \varepsilon(q_s^{k-1} | p_{(t)u}^k). \quad (\text{S14})$$

Hence, **Step 2** is implemented.

3) Implementation of Step 3

Based on (S10) and (S14), the following inferences are obtained:

$$\begin{aligned} \varepsilon(P^k, Q^{k-1}) &= \sum_{u=1}^{|U|} \sum_{t=1}^{|T|} \varepsilon(p_{(t)u}^k | q_s^{k-1}) \\ &\leq \varepsilon(P^{k-1}, Q^{k-1}) = \sum_{u=1}^{|U|} \sum_{t=1}^{|T|} \varepsilon(p_{(t)u}^{k-1} | q_s^{k-1}), \end{aligned} \quad (\text{S15})$$

$$\begin{aligned} \varepsilon(P^k, Q^k) &= \sum_{s=1}^{|S|} \sum_{t=1}^{|T|} \varepsilon(q_s^k | p_{(t)u}^k) \\ &\leq \varepsilon(P^k, Q^{k-1}) = \sum_{s=1}^{|S|} \sum_{t=1}^{|T|} \varepsilon(q_s^{k-1} | p_{(t)u}^k). \end{aligned} \quad (\text{S16})$$

Hence, it is easy to obtain the following gap:

$$\begin{aligned} &\varepsilon(P^k, Q^k) - \varepsilon(P^{k-1}, Q^{k-1}) \\ &= \varepsilon(P^k, Q^k) - \varepsilon(P^k, Q^{k-1}) \\ &\quad + \varepsilon(P^k, Q^{k-1}) - \varepsilon(P^{k-1}, Q^{k-1}). \end{aligned} \quad (\text{S17})$$

According to (S15)-(S17), we have that:

$$\varepsilon(P^k, Q^k) - \varepsilon(P^{k-1}, Q^{k-1}) \leq 0, \quad (\text{S18})$$

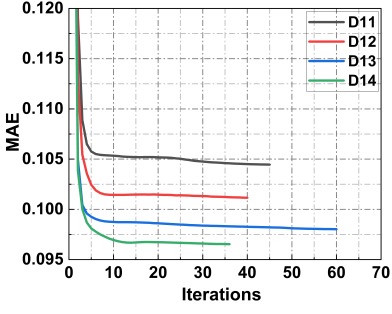
which indicates that objective function of EKL.

Hence, **Step 3** is implemented.

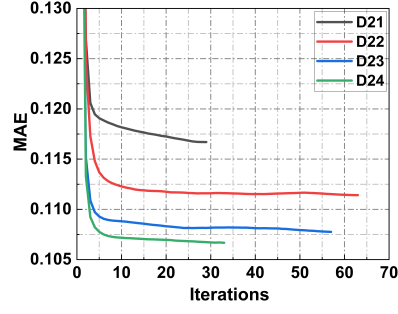
REFERENCES

- [S1] X. D. Zhang, *Matrix Analysis and Applications*. Cambridge, U.K.: Cambridge University Press, 2017.

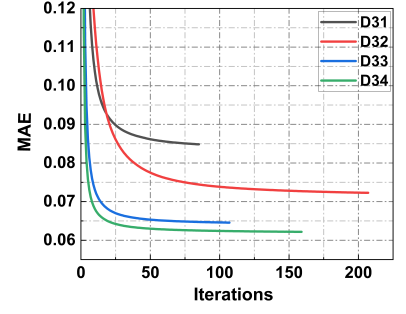
II. EXPERIMENTAL FIGURES



(a)

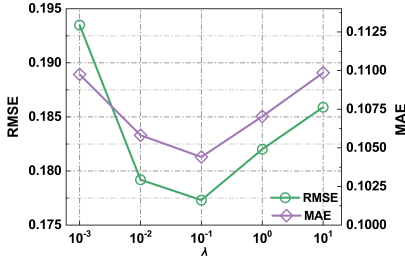


(b)

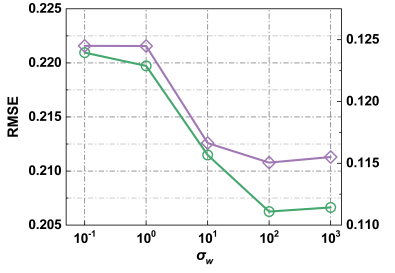


(c)

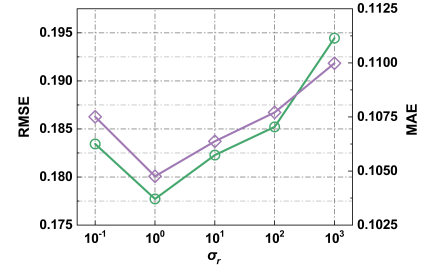
Fig. S1: Convergence curves of MAE on D1-D3.



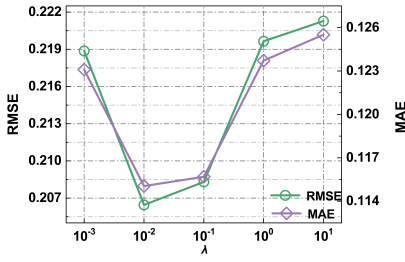
(a) Errors on D11



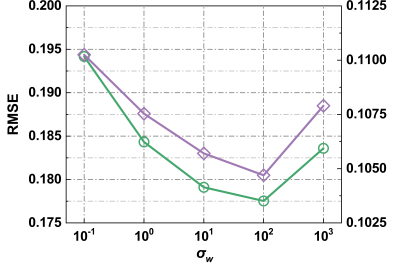
(b) Errors on D11



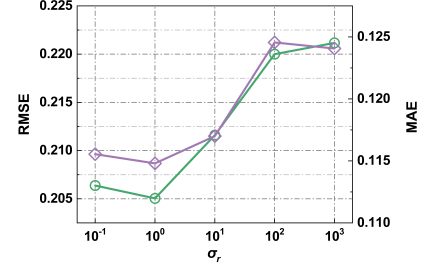
(c) Errors on D11



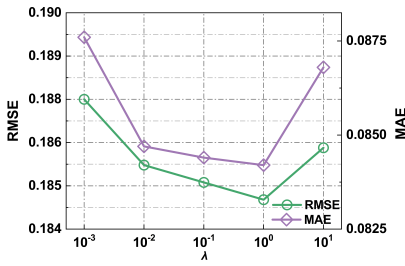
(d) Errors on D21



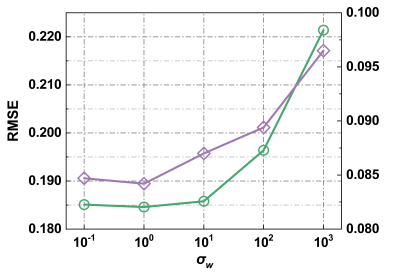
(e) Errors on D21



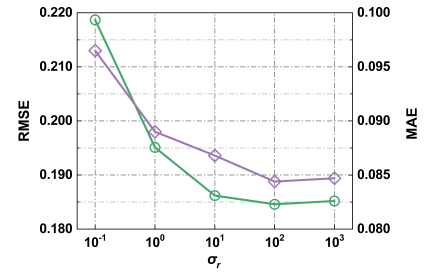
(f) Errors on D21



(g) Errors on D31



(h) Errors on D31



(i) Errors on D31

Fig. S2: Hyperparameter λ , σ_w and σ_r sensitivity analysis.

III. EXPERIMENTAL TABLES

TABLE S1: Comparison with temporal smoothing, where ★ indicates that EKL is outperformed by the compared models.

Case	Metric	EKL	CGTF	EMA-CGTF	TeDCaN	EMA-TeDCaN
D11	RMSE	0.1786 $\pm$ 9E-4	0.1821 $\pm$ 3E-4	0.1885 $\pm$ 4E-4	★ 0.1776 $\pm$ 3E-4	0.1823 $\pm$ 3E-4
	MAE	0.1053 $\pm$ 7E-4	0.1106 $\pm$ 2E-4	0.1130 $\pm$ 5E-4	0.1061 $\pm$ 6E-4	0.1097 $\pm$ 2E-4
D12	RMSE	0.1717 $\pm$ 3E-4	0.1814 $\pm$ 9E-4	0.1872 $\pm$ 6E-4	0.1737 $\pm$ 3E-4	0.1798 $\pm$ 4E-4
	MAE	0.1012 $\pm$ 2E-4	0.1098 $\pm$ 1E-4	0.1121 $\pm$ 3E-4	0.1026 $\pm$ 2E-4	0.1085 $\pm$ 3E-4
D13	RMSE	0.1667 $\pm$ 5E-4	0.1767 $\pm$ 3E-4	0.1810 $\pm$ 5E-4	0.1713 $\pm$ 4E-4	0.1755 $\pm$ 6E-4
	MAE	0.0982 $\pm$ 2E-4	0.1061 $\pm$ 3E-4	0.1094 $\pm$ 4E-4	0.1007 $\pm$ 2E-4	0.1051 $\pm$ 2E-4
D14	RMSE	0.1651 $\pm$ 4E-4	0.1739 $\pm$ 3E-3	0.1788 $\pm$ 2E-4	0.1707 $\pm$ 3E-4	0.1742 $\pm$ 4E-4
	MAE	0.0970 $\pm$ 2E-4	0.1025 $\pm$ 2E-3	0.1052 $\pm$ 3E-4	0.0999 $\pm$ 2E-4	0.1033 $\pm$ 1E-4
D21	RMSE	0.2056 $\pm$ 5E-4	0.2061 $\pm$ 2E-3	0.2108 $\pm$ 4E-4	★ 0.2021 $\pm$ 4E-4	0.2121 $\pm$ 5E-4
	MAE	0.1150 $\pm$ 5E-4	0.1190 $\pm$ 2E-3	0.1241 $\pm$ 3E-4	0.1173 $\pm$ 7E-4	0.1260 $\pm$ 4E-4
D22	RMSE	0.1984 $\pm$ 5E-4	0.2026 $\pm$ 3E-3	0.2085 $\pm$ 6E-4	★ 0.1977 $\pm$ 7E-4	0.2105 $\pm$ 4E-4
	MAE	0.1111 $\pm$ 5E-4	0.1177 $\pm$ 3E-3	0.1210 $\pm$ 2E-4	0.1135 $\pm$ 7E-4	0.1228 $\pm$ 3E-4
D23	RMSE	0.1817 $\pm$ 6E-4	0.1981 $\pm$ 8E-4	0.2033 $\pm$ 5E-4	0.1951 $\pm$ 2E-4	0.2067 $\pm$ 3E-4
	MAE	0.1070 $\pm$ 5E-4	0.1135 $\pm$ 1E-3	0.1182 $\pm$ 4E-4	0.1112 $\pm$ 2E-4	0.1195 $\pm$ 5E-4
D24	RMSE	0.1780 $\pm$ 9E-4	0.1963 $\pm$ 1E-3	0.2001 $\pm$ 4E-4	0.1930 $\pm$ 2E-4	0.2012 $\pm$ 2E-4
	MAE	0.1061 $\pm$ 5E-4	0.1113 $\pm$ 1E-3	0.1155 $\pm$ 3E-4	0.1091 $\pm$ 2E-4	0.1147 $\pm$ 4E-4
D31	RMSE	0.1845 $\pm$ 2E-03	0.2014 $\pm$ 2E-03	0.2155 $\pm$ 8E-4	0.1966 $\pm$ 2E-03	0.2108 $\pm$ 7E-4
	MAE	0.0847 $\pm$ 7E-04	0.1176 $\pm$ 1E-03	0.1284 $\pm$ 5E-4	0.1007 $\pm$ 9E-04	0.1242 $\pm$ 4E-4
D32	RMSE	0.1602 $\pm$ 8E-04	0.1879 $\pm$ 1E-03	0.1985 $\pm$ 7E-4	0.1686 $\pm$ 9E-04	0.1895 $\pm$ 5E-4
	MAE	0.0723 $\pm$ 4E-04	0.1087 $\pm$ 8E-04	0.1156 $\pm$ 3E-4	0.0892 $\pm$ 6E-04	0.1013 $\pm$ 3E-4
D33	RMSE	0.1402 $\pm$ 7E-04	0.1776 $\pm$ 9E-04	0.1844 $\pm$ 6E-4	0.1616 $\pm$ 7E-4	0.1721 $\pm$ 4E-4
	MAE	0.0644 $\pm$ 5E-04	0.1035 $\pm$ 6E-04	0.1092 $\pm$ 2E-4	0.0876 $\pm$ 4E-04	0.0915 $\pm$ 2E-4
D34	RMSE	0.1335 $\pm$ 3E-04	0.1625 $\pm$ 7E-04	0.1691 $\pm$ 5E-4	0.1549 $\pm$ 5E-04	0.1610 $\pm$ 3E-4
	MAE	0.0621 $\pm$ 3E-04	0.0945 $\pm$ 5E-04	0.0988 $\pm$ 4E-4	0.0728 $\pm$ 3E-04	0.0811 $\pm$ 2E-4

TABLE S2: Comparison on fluctuation samples, where ★ indicates that EKL is outperformed by the compared models.

Case	Metric	EKL	CGTF	TeDCaN	HRST-LR	HMLET	GTN	SGP	MGDN	PGCN	WD-GCN	SGL-ED	hetGNN-LSTM	TM-GCN	SDNMF
D11	RMSE	0.2760 $\pm$ 4E-4	0.2883 $\pm$ 2E-4	0.2832 $\pm$ 3E-4	0.4119 $\pm$ 6E-4	0.4166 $\pm$ 5E-4	0.5130 $\pm$ 8E-4	1.0814 $\pm$ 1E-3	0.6105 $\pm$ 4E-4	0.8263 $\pm$ 5E-4	0.9576 $\pm$ 3E-4	0.6837 $\pm$ 4E-4	0.9342 $\pm$ 9E-4	0.9614 $\pm$ 2E-3	0.3054 $\pm$ 4E-4
	MAE	0.1628 $\pm$ 2E-4	0.1701 $\pm$ 3E-4	0.1671 $\pm$ 2E-4	0.2430 $\pm$ 4E-4	0.2458 $\pm$ 3E-4	0.3027 $\pm$ 5E-4	0.6380 $\pm$ 9E-4	0.3602 $\pm$ 2E-4	0.4875 $\pm$ 4E-4	0.5650 $\pm$ 2E-4	0.4034 $\pm$ 3E-4	0.5512 $\pm$ 5E-4	0.5672 $\pm$ 1E-3	0.1802 $\pm$ 3E-4
D12	RMSE	0.2614 $\pm$ 3E-4	0.3297 $\pm$ 6E-4	0.2765 $\pm$ 4E-4	0.3718 $\pm$ 5E-4	0.3696 $\pm$ 4E-4	0.4533 $\pm$ 7E-4	1.0810 $\pm$ 1E-3	0.5320 $\pm$ 3E-4	0.8276 $\pm$ 6E-4	0.9568 $\pm$ 4E-4	0.5847 $\pm$ 5E-4	0.9104 $\pm$ 8E-4	0.9612 $\pm$ 2E-3	0.3006 $\pm$ 4E-4
	MAE	0.1542 $\pm$ 1E-4	0.1945 $\pm$ 4E-4	0.1631 $\pm$ 2E-4	0.2194 $\pm$ 3E-4	0.2181 $\pm$ 2E-4	0.2674 $\pm$ 4E-4	0.6378 $\pm$ 8E-4	0.3139 $\pm$ 2E-4	0.4883 $\pm$ 4E-4	0.5645 $\pm$ 3E-4	0.3450 $\pm$ 3E-4	0.5371 $\pm$ 8E-4	0.5671 $\pm$ 1E-3	0.1774 $\pm$ 2E-4
D13	RMSE	0.2519 $\pm$ 4E-4	0.2738 $\pm$ 3E-4	0.2619 $\pm$ 3E-4	0.3714 $\pm$ 4E-4	0.3150 $\pm$ 4E-4	0.4271 $\pm$ 6E-4	1.0812 $\pm$ 1E-3	0.5864 $\pm$ 4E-4	0.8243 $\pm$ 5E-4	0.9551 $\pm$ 4E-4	0.5387 $\pm$ 4E-4	0.8960 $\pm$ 7E-4	0.9619 $\pm$ 2E-3	0.2910 $\pm$ 3E-4
	MAE	0.1486 $\pm$ 2E-4	0.1615 $\pm$ 2E-4	0.1545 $\pm$ 2E-4	0.2191 $\pm$ 3E-4	0.1859 $\pm$ 2E-4	0.2520 $\pm$ 4E-4	0.6379 $\pm$ 9E-4	0.3460 $\pm$ 2E-4	0.4863 $\pm$ 3E-4	0.5635 $\pm$ 3E-4	0.3178 $\pm$ 2E-4	0.5286 $\pm$ 5E-4	0.5675 $\pm$ 1E-3	0.1717 $\pm$ 2E-4
D14	RMSE	0.2470 $\pm$ 4E-4	0.2695 $\pm$ 2E-4	0.2608 $\pm$ 3E-4	0.3711 $\pm$ 5E-4	0.2854 $\pm$ 4E-4	0.4227 $\pm$ 7E-4	1.0812 $\pm$ 1E-3	0.5350 $\pm$ 3E-4	0.8216 $\pm$ 5E-4	0.9583 $\pm$ 4E-4	0.5316 $\pm$ 3E-4	0.8861 $\pm$ 8E-4	0.9621 $\pm$ 2E-3	0.2867 $\pm$ 4E-4
	MAE	0.1457 $\pm$ 1E-4	0.1590 $\pm$ 2E-4	0.1539 $\pm$ 2E-4	0.2190 $\pm$ 3E-4	0.1684 $\pm$ 2E-4	0.2494 $\pm$ 4E-4	0.6379 $\pm$ 8E-4	0.3157 $\pm$ 2E-4	0.4844 $\pm$ 4E-4	0.5654 $\pm$ 2E-4	0.3136 $\pm$ 2E-4	0.5228 $\pm$ 4E-4	0.5676 $\pm$ 1E-3	0.1691 $\pm$ 2E-4
D21	RMSE	0.3527 $\pm$ 5E-4	0.3581 $\pm$ 8E-4	0.3615 $\pm$ 6E-4	0.3640 $\pm$ 4E-4	0.4848 $\pm$ 5E-4	0.5155 $\pm$ 7E-4	0.8236 $\pm$ 9E-4	0.5622 $\pm$ 4E-4	0.6984 $\pm$ 6E-4	0.7408 $\pm$ 5E-4	0.6346 $\pm$ 5E-4	0.6845 $\pm$ 6E-4	0.7452 $\pm$ 1E-3	0.3980 $\pm$ 5E-4
	MAE	0.1975 $\pm$ 3E-4	0.2005 $\pm$ 4E-4	0.2024 $\pm$ 4E-4	0.2038 $\pm$ 2E-4	0.2715 $\pm$ 3E-4	0.2887 $\pm$ 4E-4	0.4612 $\pm$ 5E-4	0.3148 $\pm$ 2E-4	0.3911 $\pm$ 3E-4	0.4148 $\pm$ 3E-4	0.3554 $\pm$ 3E-4	0.3833 $\pm$ 4E-4	0.4173 $\pm$ 7E-4	0.2228 $\pm$ 3E-4
D22	RMSE	0.3471 $\pm$ 4E-4	0.3578 $\pm$ 7E-4	0.3552 $\pm$ 5E-4	0.3601 $\pm$ 3E-4	0.4363 $\pm$ 6E-4	0.4856 $\pm$ 8E-4	0.8234 $\pm$ 9E-4	0.4993 $\pm$ 4E-4	0.6991 $\pm$ 5E-4	0.7407 $\pm$ 4E-4	0.5643 $\pm$ 5E-4	0.6789 $\pm$ 6E-4	0.7444 $\pm$ 1E-3	0.3905 $\pm$ 4E-4
	MAE	0.1944 $\pm$ 2E-4	0.2004 $\pm$ 4E-4	0.1989 $\pm$ 3E-4	0.2017 $\pm$ 2E-4	0.2443 $\pm$ 3E-4	0.2719 $\pm$ 4E-4	0.4611 $\pm$ 6E-4	0.2796 $\pm$ 2E-4	0.3915 $\pm$ 3E-4	0.4148 $\pm$ 3E-4	0.3160 $\pm$ 2E-4	0.3802 $\pm$ 3E-4	0.4169 $\pm$ 6E-4	0.2187 $\pm$ 2E-4
D23	RMSE	0.3334 $\pm$ 5E-4	0.3568 $\pm$ 6E-4	0.3436 $\pm$ 4E-4	0.3442 $\pm$ 4E-4	0.3932 $\pm$ 5E-4	0.4691 $\pm$ 7E-4	0.8236 $\pm$ 9E-4	0.4664 $\pm$ 3E-4	0.6991 $\pm$ 5E-4	0.7408 $\pm$ 4E-4	0.5382 $\pm$ 5E-4	0.6514 $\pm$ 6E-4	0.7443 $\pm$ 1E-3	0.3787 $\pm$ 4E-4
	MAE	0.1867 $\pm$ 3E-4	0.1998 $\pm$ 4E-4	0.1924 $\pm$ 3E-4	0.1928 $\pm$ 2E-4	0.2202 $\pm$ 3E-4	0.2627 $\pm$ 4E-4	0.4612 $\pm$ 5E-4	0.2612 $\pm$ 2E-4	0.3915 $\pm$ 3E-4	0.4148 $\pm$ 3E-4	0.3014 $\pm$ 2E-4	0.3648 $\pm$ 3E-4	0.4168 $\pm$ 6E-4	0.2120 $\pm$ 2E-4
D24	RMSE	0.3321 $\pm$ 4E-4	0.3556 $\pm$ 6E-4	0.3431 $\pm$ 4E-4	0.3444 $\pm$ 4E-4	0.3654 $\pm$ 4E-4	0.4670 $\pm$ 7E-4	0.8238 $\pm$ 9E-4	0.4567 $\pm$ 3E-4	0.6946 $\pm$ 5E-4	0.7408 $\pm$ 4E-4	0.4701 $\pm$ 5E-4	0.7025 $\pm$ 7E-4	0.7423 $\pm$ 1E-3	0.3755 $\pm$ 5E-4
	MAE	0.1860 $\pm$ 2E-4	0.1991 $\pm$ 4E-4	0.1921 $\pm$ 3E-4	0.1929 $\pm$ 2E-4	0.2046 $\pm$ 3E-4	0.2615 $\pm$ 4E-4	0.4613 $\pm$ 6E-4	0.2558 $\pm$ 2E-4	0.3890 $\pm$ 3E-4	0.4148 $\pm$ 3E-4	0.2633 $\pm$ 2E-4	0.3934 $\pm$ 4E-4	0.4157 $\pm$ 6E-4	0.2102 $\pm$ 3E-4
D31	RMSE	0.3044 $\pm$ 9E-4	0.3725 $\pm$ 1E-3	0.3440 $\pm$ 7E-4	0.3226 $\pm$ 5E-4	0.4113 $\pm$ 4E-4	1.1808 $\pm$ 2E-3	0.8588 $\pm$ 1E-3	1.2876 $\pm$ 2E-3	0.7641 $\pm$ 8E-4	0.7293 $\pm$ 7E-4	1.2878 $\pm$ 2E-3	0.7410 $\pm$ 9E-4	0.8492 $\pm$ 1E-3	0.4901 $\pm$ 8E-4
	MAE	0.1400 $\pm$ 4E-4	0.1714 $\pm$ 6E-4	0.1582 $\pm$ 4E-4	0.1484 $\pm$ 2E-4	0.1892 $\pm$ 2E-4	0.5432 $\pm$ 9E-4	0.3950 $\pm$ 7E-4	0.5923 $\pm$ 8E-4	0.3515 $\pm$ 4E-4	0.3355 $\pm$ 3E-4	0.5924 $\pm$ 1E-3	0.3409 $\pm$ 4E-4	0.3906 $\pm$ 5E-4	0.2254 $\pm$ 4E-4
D32	RMSE	0.2643 $\pm$ 6E-4	0.3476 $\pm$ 9E-4	0.2950 $\pm$ 5E-4	0.2890 $\pm$ 4E-4	0.4045 $\pm$ 5E-4	1.1224 $\pm$ 2E-3	0.9646 $\pm$ 1E-3	1.2864 $\pm$ 2E-3	0.7644 $\pm$ 7E-4	0.7398 $\pm$ 6E-4	1.2870 $\pm$ 2E-3	0.7326 $\pm$ 8E-4	0.8762 $\pm$ 1E-3	0.4087 $\pm$ 5E-4
	MAE	0.1216 $\pm$ 3E-4	0.1599 $\pm$ 5E-4	0.1357 $\pm$ 3E-4	0.1329 $\pm$ 2E-4	0.1861 $\pm$ 2E-4	0.5163 $\pm$ 8E-4	0.4437 $\pm$ 7E-4	0.5917 $\pm$ 9E-4	0.3516 $\pm$ 4E-4	0.3403 $\pm$ 3E-4	0.5920 $\pm$ 1E-3	0.3370 $\pm$ 4E-4	0.4031 $\pm$ 5E-4	0.1879 $\pm$ 3E-4
D33	RMSE	0.2313 $\pm$ 5E-4	0.3285 $\pm$ 8E-4	0.2828 $\pm$ 4E-4	0.2860 $\pm$ 4E-4	0.4014 $\pm$ 5E-4	0.9550 $\pm$ 1E-3	0.8160 $\pm$ 9E-4	1.2775 $\pm$ 2E-3	0.7443 $\pm$ 7E-4	0.7419 $\pm$ 6E-4	1.2688 $\pm$ 2E-3	0.7056 $\pm$ 8E-4	0.9102 $\pm$ 1E-3	0.3584 $\pm$ 5E-4
	MAE	0.1064 $\pm$ 2E-4	0.1511 $\pm$ 4E-4	0.1301 $\pm$ 3E-4	0.1316 $\pm$ 2E-4	0.1846 $\pm$ 2E-4	0.4393 $\pm$ 7E-4	0.3754 $\pm$ 6E-4	0.5877 $\pm$ 8E-4	0.3424 $\pm$ 4E-4	0.3413 $\pm$ 3E-4	0.5836 $\pm$ 1E-3	0.3246 $\pm$ 4E-4	0.4187 $\pm$ 5E-4	0.1648 $\pm$ 3E-4
D34	RMSE	0.2202 $\pm$ 4E-4	0.3006 $\pm$ 7E-4	0.2710 $\pm$ 4E-4	0.2853 $\pm$ 3E-4	0.4014 $\pm$ 5E-4	0.8420 $\pm$ 1E-3	0.8225 $\pm$ 9E-4	1.2443 $\pm$ 2E-3	0.7479 $\pm$ 6E-4	0.7324 $\pm$ 6E-4	1.2353 $\pm$ 2E-3	0.6546 $\pm$ 7E-4	0.9275 $\pm$ 1E-3	0.3370 $\pm$ 4E-4
	MAE	0.1013 $\pm$ 2E-4	0.1383 $\pm$ 4E-4	0.1247 $\pm$ 3E-4	0.1312 $\pm$ 2E-4	0.1846 $\pm$ 2E-4	0.3873 $\pm$ 6E-4	0.3784 $\pm$ 6E-4	0.5724 $\pm$ 8E-4	0.3440 $\pm$ 4E-4	0.3369 $\pm$ 3E-4	0.5682 $\pm$ 1E-3	0.3011 $\pm$ 4E-4	0.4267 $\pm$ 5E-4	0.1550 $\pm$ 3E-4
Loss/Win	—	24/0	24/0	24/0	24/0	24/0	24/0	24/0	24/0	24/0	24/0	24/0	24/0	24/0	24/0
Rank	1.00	3.50	2.25	3.92	5.42	8.75	12.83	9.83	9.92	10.58	10.33	9.50	12.25	4.92	4.92
p-value	—	1.82E-05	1.19E-07	1.19E-07	1.19E-07	1.19E-07	1.19E-07	1.82E-05	1.19E-07	1.19E-07	1.19E-07	1.19E-07	1.19E-07	1.19E-07	1.19E-07

TABLE S3: EKL performance with temporal split setting.

Dataset	Model	RMSE	MAE
D1	EKL	0.1636 $\pm$ 3E-4	0.